REPORT OF INNOVATIVE PROJECT I

Project – I

Bachelor of Technology

in

Computer Science and Engineering

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ABSTRACT

This project presents the implementation of the K-Means clustering algorithm from scratch using Python, focusing on unsupervised machine learning techniques for data analysis. The primary objective of the project is to group similar data points into clusters based on their features without relying on predefined labels. The algorithm initializes centroids randomly and iteratively refines them by assigning data points to the nearest cluster and updating centroid positions until convergence is achieved.

The implementation emphasizes core concepts such as distance calculation using Euclidean distance, cluster assignment, and centroid optimization. Additionally, the model handles edge cases such as empty clusters by reinitializing centroids, ensuring robustness and stability. The algorithm is tested on a dataset of student data, and the clustering results are visualized using scatter plots to demonstrate the effectiveness of grouping.

This project enhances understanding of clustering techniques, algorithm design, and data visualization. It also provides practical insights into how machine learning models work internally without relying on external libraries like Scikit-learn. The outcome of this project demonstrates the applicability of K-Means clustering in real-world data analysis and pattern recognition tasks.

The program is implemented in Python and is designed to perform clustering using the K-Means algorithm developed from scratch. It takes as input a dataset containing multiple data points and groups them into a predefined number of clusters based on similarity. The algorithm begins by randomly initializing centroids and iteratively assigns each data point to

the nearest centroid using Euclidean distance. The centroids are then recalculated as the mean of the assigned points, and the process continues until convergence is achieved or a maximum number of iterations is reached.

For each iteration, the program updates cluster assignments and centroid positions, effectively demonstrating how the grouping evolves over time. The final clustering results are visualized using scatter plots, providing a clear representation of how data points are partitioned into distinct groups. Special care is taken to handle edge cases such as empty clusters, where centroids are reinitialized to ensure the algorithm remains stable and effective.

The program further illustrates how different initializations and cluster numbers can influence the final outcome, offering insights into the sensitivity and behavior of the K-Means algorithm. By applying the model to a student dataset, the project highlights practical applications of clustering in identifying patterns and grouping similar entities.

Overall, this project demonstrates the fundamental concepts of unsupervised learning, iterative optimization, and pattern recognition through a computational approach. The implementation emphasizes a deeper understanding of how clustering algorithms function internally, without relying on external libraries, and showcases the importance of such techniques in data analysis, machine learning, and real-world problem-solving.

1. Introduction :

In recent years, the rapid growth of data has made it essential to develop efficient techniques for extracting meaningful patterns and insights. Clustering is one of the most important unsupervised machine learning methods used to group similar data points without prior knowledge of labels. It plays a crucial role in various real-world applications such as market segmentation, recommendation systems, geographical analysis, and anomaly detection.

This project focuses on the implementation and analysis of two widely used clustering algorithms: **K-Means** and **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)**. K-Means is a centroid-based algorithm that partitions data into a predefined number of clusters by minimizing the distance between data points and cluster centers. It is simple, efficient, and works well for datasets with clearly separable spherical clusters. However, it requires prior knowledge of the number of clusters and is sensitive to outliers.

On the other hand, DBSCAN is a density-based clustering algorithm that groups data points based on their density distribution. Unlike K-Means, it does not require specifying the number of clusters in advance and can effectively identify clusters of arbitrary shapes while also detecting noise or outliers. This makes DBSCAN particularly useful for real-world datasets where clusters are irregular and noisy.

In this project, both algorithms are applied to multiple datasets, including geographical data (latitude and longitude) and student performance data (CGPA and IQ). The performance of the models is evaluated using techniques such as the Elbow Method and silhouette score, and the results are visualized using scatter plots for better interpretation.

The aim of this project is to understand the working principles, advantages, and limitations of different clustering techniques, and to demonstrate how they can be used to uncover hidden patterns in data. Through this comparative analysis, the project highlights the importance of choosing appropriate algorithms and parameters for different types of datasets.

2. What is Clustering:

Clustering is an unsupervised machine learning technique used to group similar data points together based on their features or characteristics. In clustering, no predefined labels are given; instead, the algorithm identifies patterns and organizes the data into meaningful groups called clusters. Data points within the same cluster are more similar to each other than to those in different clusters. Clustering is widely used in applications such as customer segmentation, pattern recognition, and data analysis.

3. Types of Clustering

Clustering techniques can be broadly categorized based on how they group data points. In this project, two important types of clustering methods are used: **centroid-based clustering** and **density-based clustering**.

Centroid-based clustering (K-Means):

Centroid-based clustering methods group data points based on their distance from a central point called a centroid. The K-Means algorithm is one of the most widely used approaches in this category. It partitions the dataset into a predefined number of clusters (K) by minimizing the distance between data points and their respective cluster centroids. In this project, K-Means is applied to geographical data (latitude and longitude) as well as student data (CGPA and IQ). The Elbow Method is

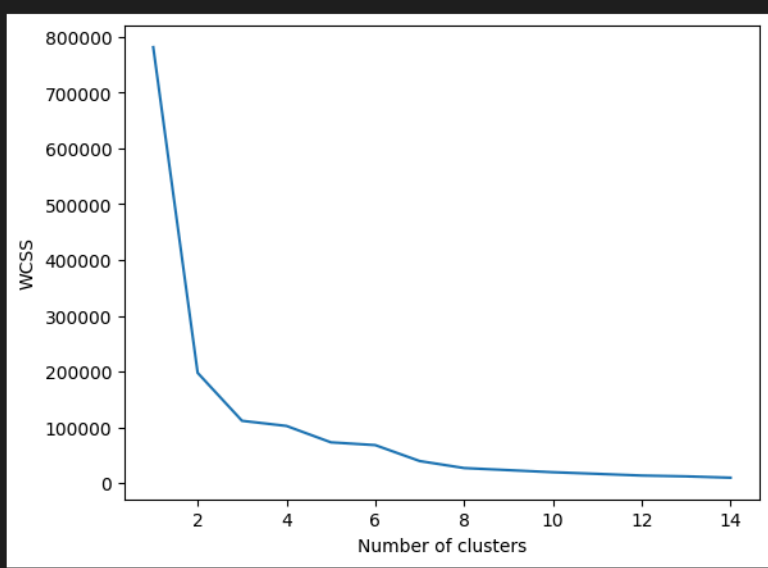
used to determine the optimal number of clusters, and the results are evaluated using the silhouette score (approximately 0.52), indicating moderate clustering performance. K-Means works efficiently for well-separated and spherical clusters but is sensitive to outliers and requires the number of clusters to be specified in advance.

```
wcss = []

for i in range(1, 15):
    KM = KMeans(n_clusters=i, random_state=42)
    KM.fit(df[['latitude', 'longitude']])
    wcss.append(KM.inertia_)
```

```
plt.plot(range(1, 15), wcss)
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
```

```
Text(0, 0.5, 'WCSS')
```



Density-based clustering (DBSCAN):

Density-based clustering methods group data points based on the density of data regions rather than distance from a centroid. DBSCAN identifies clusters as areas of high density separated by regions of low density. It uses parameters such as epsilon (ε) and minimum samples to define clusters. One of the key advantages of DBSCAN is its ability to detect noise or outliers and to identify clusters of arbitrary shapes. In this project, DBSCAN is applied to both synthetic and real-world datasets.

For the student dataset, it produces well-defined clusters with a high silhouette score (approximately 0.86), indicating strong clustering quality. However, when applied to geographical data, the performance varies (silhouette score around 0.44), highlighting its sensitivity to parameter selection. Overall, DBSCAN is more flexible than K-Means and is particularly useful for datasets with irregular structures and noise.

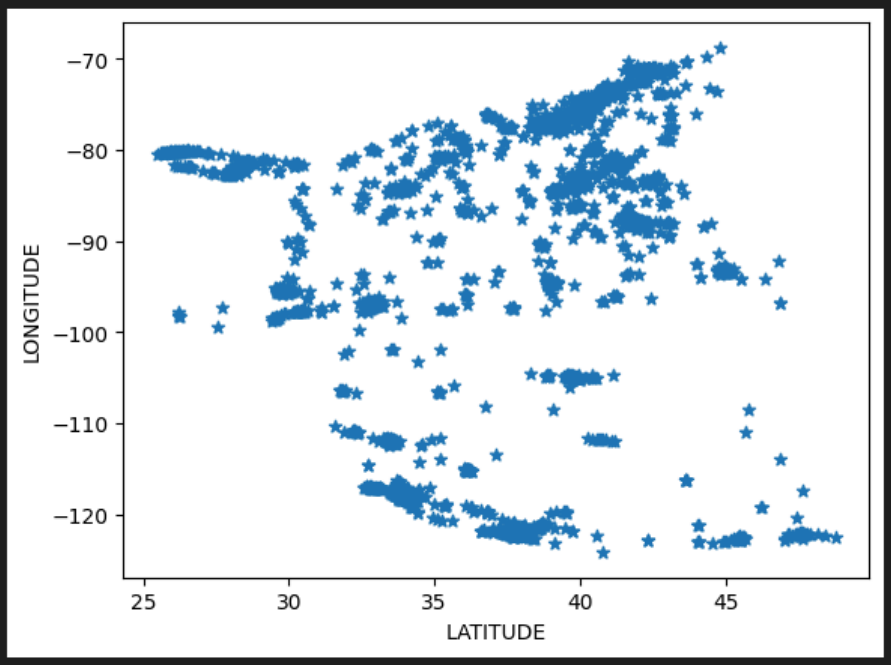
4. Algorithms Used in Your Project

K-Means Algorithm

K-Means is a centroid-based clustering algorithm that partitions the dataset into a predefined number of clusters (K). It works by randomly initializing centroids and assigning each data point to the nearest centroid using distance measures such as Euclidean distance. The centroids are then updated as the mean of the assigned points, and the process repeats until convergence. In this project, K-Means is applied to datasets like geographical coordinates and student data, and the optimal number of clusters is determined using the Elbow Method. It is efficient and easy to implement but requires prior knowledge of the number of clusters and is sensitive to outliers.

```
import matplotlib.pyplot as plt
plt.scatter(df['latitude'], df['longitude'], marker="*")
plt.xlabel('LATITUDE')
plt.ylabel('LONGITUDE')
```

```
Text(0, 0.5, 'LONGITUDE')
```



DBSCAN Algorithm

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that groups data points based on the density of regions. It uses two main parameters: epsilon (ε), which defines the neighborhood radius, and minimum samples, which specifies the minimum number of points required to form a dense region. Points in high-density areas are grouped into clusters, while low-density points are treated as noise or outliers. In this project, DBSCAN is used on both synthetic and real datasets, effectively identifying clusters of arbitrary shapes and detecting noise. Unlike K-Means, it does not require the number of clusters in advance but is sensitive to parameter selection.

5. Datasets Used

In this project, two different types of datasets are used to evaluate the performance of clustering algorithms under varying conditions.

Geographical Dataset (Chipotle Stores):

This dataset consists of location-based data containing the **latitude and longitude** of Chipotle store locations. It is used to analyze spatial clustering and identify geographical groupings of stores. The dataset exhibits varying density across regions, making it suitable for testing both K-Means and DBSCAN algorithms. K-Means groups the locations into clusters based on distance, while DBSCAN identifies dense regions and detects sparse areas as noise. The results show that clustering performance depends on the distribution of points, with moderate silhouette scores (around 0.44–0.52) indicating partially defined clusters.

state	location	address	latitude	longitude
Alabama	Auburn	346 W Magnolia Ave	32.60681297	-85.48732833
Alabama	Birmingham	300 20th St S Birmir	33.5097215	-86.80275567
Alabama	Birmingham	3220 Morrow Rd Bim	33.59558141	-86.64743685
Alabama	Birmingham	4719 Highway 280 B	33.42258215	-86.69827947
Alabama	Cullman	1821 Cherokee Ave S	34.15413377	-86.84122008
Alabama	Hoover	1759 Montgomery H	33.37895803	-86.8038021
Alabama	Huntsville	5900 University Dr N	34.74231925	-86.66572046
Alabama	Mobile	3871 Airport Blvd Mc	30.67533781	-88.14375393
Alabama	Mobile	7765 Airport Blvd D1	30.68273058	-88.22499816
Alabama	Montgomery	2560 Berryhill Rd St	32.35917688	-86.16225285
Alabama	Opelika	2125 Interstate Dr Op	32.61680875	-85.40447877
Alabama	Prattville	2566 Cobbs Ford Rd	32.459169	-86.391343

Student Dataset (CGPA and IQ):

This dataset contains student-related features such as **CGPA and IQ**, representing academic performance and cognitive ability. It is used to identify patterns and group students with similar characteristics. After preprocessing and standardization, clustering is applied using both K-Means and DBSCAN. The results show clearly defined clusters, especially with DBSCAN, which achieves a high silhouette score (approximately 0.86), indicating strong separation between groups. This dataset demonstrates how clustering can be effectively used for analyzing structured and well-distributed data.

cgpa	iq
5.13	88
5.9	113
8.36	93
8.27	97
5.45	110
5.88	109
8.41	98

6. Evaluation Methods:

To evaluate the performance of clustering algorithms used in this project, the **Silhouette Score** is used as a key metric. It measures how well data points are grouped within clusters and how distinct the clusters are from each other.

Silhouette Score Formula

$$s = \frac{b - a}{\max(a, b)}$$

where:

- a = average intra-cluster distance
- b = average nearest-cluster distance

Range:

- **+1** → Strong clustering
- **0** → Overlapping clusters
- **-1** → Poor clustering

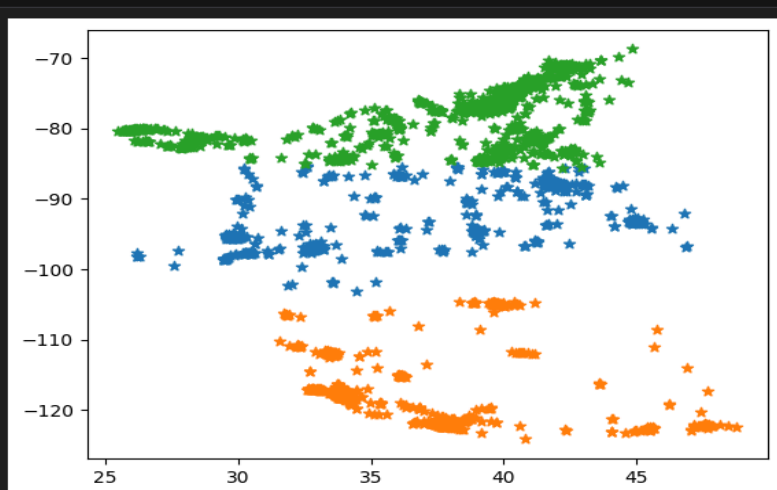
Final Results from the Project

1. K-Means (Geographical Dataset)

- **Silhouette Score: 0.5214**
- Indicates **moderate clustering**
- Clusters are reasonably separated but not perfect
- Some overlap exists due to uneven spatial distribution

```
for i in range(3):
    plt.scatter(X[y_means == i, 0], X[y_means == i, 1], marker='*')

# plt.legend(['cluster 0', 'cluster 1', 'cluster 2'])
```



Generate + Code + Markdown

```
from sklearn.metrics import silhouette_score

silhouette_score(X, y_means)
```

```
0.5214330201421082
```

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2. DBSCAN (Geographical Dataset – Updated Result)

- **Silhouette Score: 0.2123**
- Indicates **weak clustering performance**
- Poor separation between clusters
- Many points grouped into a single cluster or treated as noise
- Shows high sensitivity to parameters like ε\varepsilonε and **min_samples**

Comparison of Results

Algorithm	Dataset Type	Silhouette Score	Performance
DBSCAN	Location Data	**0.2123**	Weak
K-Means	Location Data	**0.5214**	Moderate

Analysis

- **Best Performance:**
  K-Means on geographical data → stable but limited
- **Worst Performance:**
  DBSCAN on geographical data → poor due to:
  - uneven density
  - improper parameter tuning
  - large variation in spatial distribution

7. Purpose

K-Means Clustering (Based on Project Results)

In this project, the K-Means clustering algorithm is applied to the **Chipotle stores dataset**, which contains geographical features such as **latitude and longitude**. The goal is to group store locations into clusters based on their spatial proximity.

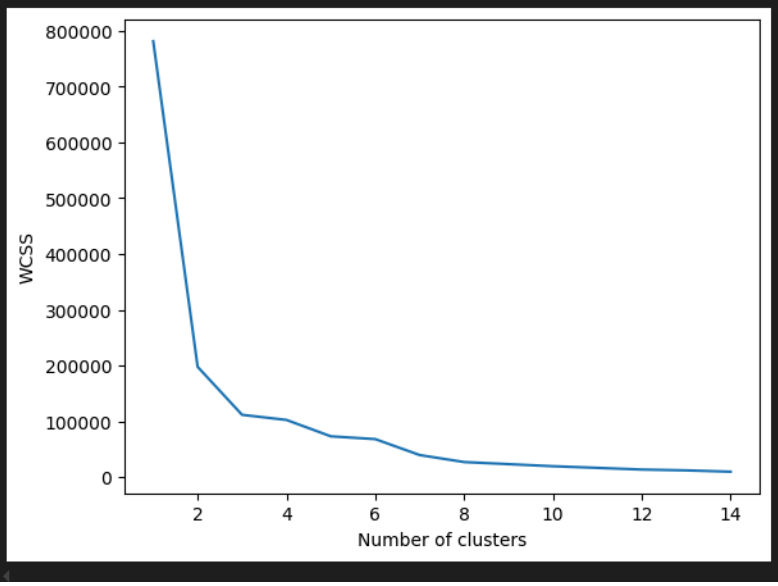
Initially, the dataset is visualized using a scatter plot, which shows the distribution of store locations across different regions. Since K-Means requires the number of clusters to be specified in advance, the **Elbow Method** is used to determine the optimal value of KKK. This is done by calculating the **Within-Cluster Sum of Squares (WCSS)** for different values of KKK (from 1 to 14) and plotting the results.

```
wcss = []

for i in range(1, 15):
    KM = KMeans(n_clusters=i, random_state=42)
    KM.fit(df[['latitude', 'longitude']])
    wcss.append(KM.inertia_)
```

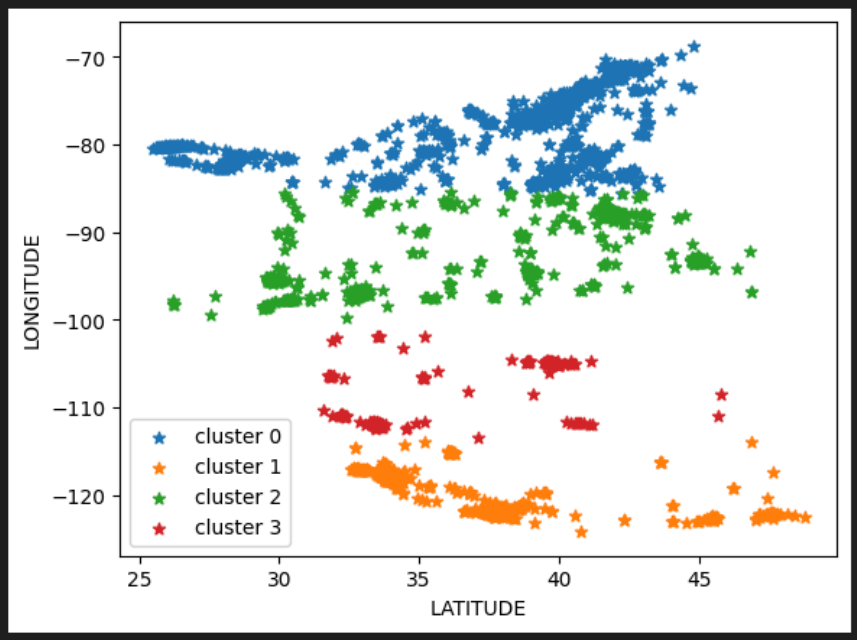
```
plt.plot(range(1, 15), wcss)
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
```

```
Text(0, 0.5, 'WCSS')
```



```python
for i in range(4):
    plt.scatter(X[y_means == i, 0], X[y_means == i, 1], marker='*')

plt.legend(['cluster 0', 'cluster 1', 'cluster 2', 'cluster 3'])
plt.xlabel('LATITUDE')
plt.ylabel('LONGITUDE')
plt.show()
```



What is WCSS?

WCSS is calculated as:

WCSS=∑(distance between data point and its centroid)^2

- It measures **cluster compactness**
- Lower value → tighter clusters

From the Elbow graph, a noticeable bend is observed around **K=3K = 3K=3 or K=4K = 4K=4**, indicating that these values provide a good balance between model complexity and clustering accuracy. Based on this, the algorithm is implemented with **K=4K = 4K=4** clusters.

After applying K-Means, the dataset is divided into four distinct clusters. The results are visualized using a scatter plot with

different colors representing each cluster. The clusters show meaningful geographical segmentation of store locations, where nearby points are grouped together.

To evaluate the clustering performance, the **Silhouette Score** is calculated:

- **Silhouette Score (K-Means): 0.5214**

This value indicates **moderate clustering quality**, meaning that:

- Clusters are reasonably well-separated
- Some overlap exists between clusters
- The model captures general patterns but is not perfectly accurate

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8.Conclusion (K-Means Analysis)

The results demonstrate that K-Means is effective in grouping geographical data into clusters based on distance. It provides a clear and interpretable segmentation of locations. However, its performance is limited by assumptions such as:

- Fixed number of clusters
- Sensitivity to data distribution
- Difficulty handling irregular cluster shapes

Despite these limitations, K-Means proves to be a simple, efficient, and useful algorithm for clustering structured data in this project.